

① `int sequentialSearch(int n, int* A, int x) {`
`int i = 0;`
`while (i < n && A[i] != x)` Basic operation
`i++;`
`if (i < n)`
`return i;`
`else`
`return -1;`
`}`

$C_{best}(n) = 1$ $C_{worst}(n) = \sum_{i=0}^{n-1} 1 \rightarrow n-1 \approx n$

$C_{avg}(n) \rightarrow p: 0 \leq p \leq 1$ i.e. poz.сында olmasi $\frac{p}{n}$ olasılık

$$C_{avg}(n) = n \cdot (1-p) \left[\frac{p}{n} + \frac{2p}{n} + \frac{3p}{n} \dots \frac{np}{n} \right]$$

$\downarrow$
eger yoksa

varsa

$$\sum_{i=1}^n \frac{p}{n} \cdot i = \frac{n \cdot (n+1) \cdot p}{2n} \rightarrow \frac{(n+1)p}{2}$$

$\rightarrow p=0$ için $C_{avg}(n) = n + 0 = n$

$p=1$ için $C_{avg}(n) = 0 + \frac{n+1}{2} \approx n$

$C_{avg}(n) = n \rightarrow O(n)$

$C_{worst}(n) = n \rightarrow O(n)$

$C_{best}(n) = 1 \rightarrow \Omega(1)$

② $\frac{1}{2}n(n-1) \in \Theta(n^2)$

if $0 \leq c_2 * g(n) \leq f(n) \leq c_1 * g(n)$ and $c_1 \geq c_2, c_1 > 0, c_2 > 0$

then $f(n) \in \Theta(g(n))$

$$f(n) = \frac{1}{2}n^2 - \frac{1}{2}n$$

$$c_2 = \frac{1}{3} \rightarrow \frac{1}{3}n^2 \leq \frac{1}{2}n^2 - \frac{1}{2}n \rightarrow n_0 = 25$$

$n_0 = 25$ $c_2 = \frac{1}{3} \rightarrow \Omega(n^2)$

$$c_1 = 15 \rightarrow \frac{1}{2}n^2 - \frac{1}{2}n \leq 15n^2 \rightarrow n_0 = 25$$

$n_0 = 25$ $c_1 = 15 \rightarrow O(n^2)$

$\Theta(n^2)$

$15 > \frac{1}{3}$

$15 > 0$
 $\frac{1}{3} > 0$

$$(3) a. \sum_{i=3}^{n+1} i \rightarrow (n+1-3+1) \cdot \frac{(n+1+3)}{2} = (n-1) \frac{(n+4)}{2} = \frac{n^2+3n-4}{2}$$

$$b. \sum_{i=0}^{n-1} i(i+1) \rightarrow \sum_{i=0}^{n-1} i^2 + \sum_{i=0}^{n-1} i$$

$$\downarrow \quad \downarrow$$

$$\frac{(n-1)(n)(2n-1)}{6} + \frac{(n-1)(n)}{2} \rightarrow \frac{(2n^2-n)(n-1) + (n-1)(n) \cdot 3}{6}$$

$$> \frac{(n-1)(2n^2+2n)}{6}$$

$$(4) x(n/2) = x(n/4) + n/2 \quad x(n/4) = x(n/8) + n/4 \rightarrow x(n) = x(n/8) + \frac{7n}{4}$$

$$x(n/2) = x(n/8) + \frac{3n}{2} \rightarrow x(n) = x\left(\frac{n}{2^k}\right) + \frac{(2^k-1)n}{2^{k-1}}$$

$$\rightarrow n = 2^k \quad k = \log_2 n$$

$$x(n) = x\left(\frac{2^k}{2^k}\right) + \frac{(2^{\log_2 n} - 1)n}{2} \rightarrow \frac{(n-1)n}{2} + 1 \rightarrow 2n-2+1 = 2n-1$$

(5) dToDecimal(m, number, d)

```

digit ← 1
sum ← 0
for i ← 0 to m do
    endDigit ← number mod 10
    sum ← sum + endDigit * digit
    digit ← d * digit
    number ← number / 10
return sum

```

$m+1$ control
basic operation
 $k \neq m$ operation

$$O \rightarrow 5m+1 \leq c_1 k m \rightarrow c_1 = 20 \rightarrow O(n)$$

$$n_0 = 1$$

$$\Omega \rightarrow c_2 k m \leq 5m+1 \quad c_2 = 1 \rightarrow \Omega(n)$$

$$n_0 = 1$$

$$c_1 \geq c_2 \quad c_1 > 0 \quad c_2 > 0 \quad n_0 = 1$$

$$O \leq c_2 k m \leq 5m+1 \leq c_1 k m \rightarrow \Theta(n)$$

DAC (A[0...n-1], low, high)

if high $\leq$ low then
return A[low]

mid = (low + high) / 2

sum = 0; left_max = min(A);
for i = mid; i $\geq$ low; i--

sum = sum + A[i];
if (sum > left_max)
left_max = sum;

right_max = min(A); sum = 0;

for i = mid + 1; i $\leq$ high; i++

sum = sum + A[i];
if (sum > right_max)
right_max = sum;

left-right = max (DAC (A[], low, mid), DAC (A[], mid + 1, high))

return max (left-right, left_max + right_max)

DIVIDE

AND

CONQUER

$$T(n) = 2T(n/2) + N \rightarrow \begin{matrix} d=1 \\ a=2 \\ b=2 \end{matrix} \quad 2 = 2^1 \rightarrow n^1 \log n \rightarrow n \log n$$

Her durumda tüm elemanları değişim 2 erunde olduğu için

$$C_{best} = \Theta(n \log n) \quad C_{average} = \Theta(n \log n) \quad C_{worst} = \Theta(n \log n)$$

BF(N, A[N])

max = A[0]

for i=0; i < N; i=i+1

tmp=0

for j=i; j < N; j=j+1

tmp = tmp + A[j]

if (tmp > max)

max = tmp

left = i;

right = j;

return max

BRUTE

FORCE

$\rightarrow N$ defa dış döngü

$\rightarrow N-i$ defa iç döngü

$$N + (N-1) + (N-2) \dots 1 \rightarrow \frac{N * (N+1)}{2} \rightarrow N^2$$

Her durumda tüm elemanları kontrol edeceği için

$$C_{best} = \Theta(n \log n)$$

$$C_{arrange} = \Theta(n \log n)$$

$$C_{wrt} = \Theta(n \log n)$$